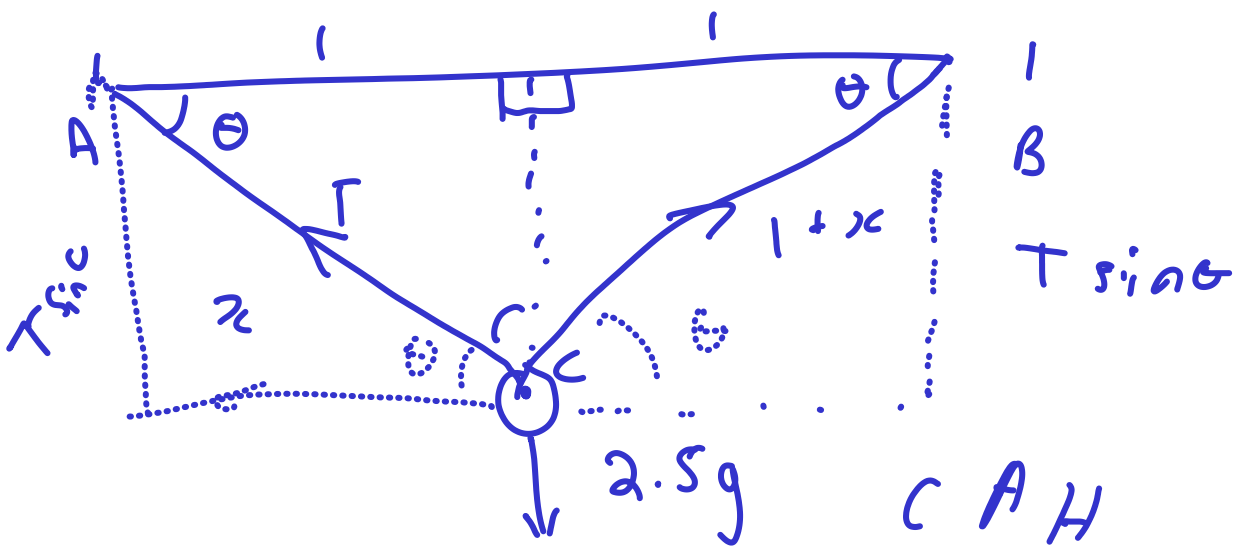
1. An elastic string of natural length $2\ m$ and modulus of elasticity $40\ N$ is stretched between two points A and B on the same horizontal level, where the distance AB is $2\ m$. A particle of mass $2.5\ kg$ is attached to the midpoint of the string. Find the angle the two parts of the string make with the horizontal AB. (Set up the equation and solve).



C A H

R (↑)

$2T\sin\theta - 2.5g = 0$

Nat length of AC = 1

length of AC = $\frac{1}{\cos\theta}$

Extension = $\frac{1}{\cos\theta} - 1$

$\lambda = 40\ N$

$T = \frac{\lambda x}{1}$

$T = 40\left(\frac{1}{\cos\theta} - 1\right)$

Sub into ①

$2T\sin\theta - 5g = 0$

$2(40)\left(\frac{1}{\cos\theta} - 1\right)\sin\theta - 2.0g = 0$

$80\left(\frac{1}{\cos\theta} - 1\right)\sin\theta - 2.5g = 0$

$80\tan\theta - 80\sin\theta - 2.5g = 0$

$80\tan\theta - 80\sin\theta = 2.5g$

$80\tan\theta - 80\sin\theta = 24.5$

Solve using calc

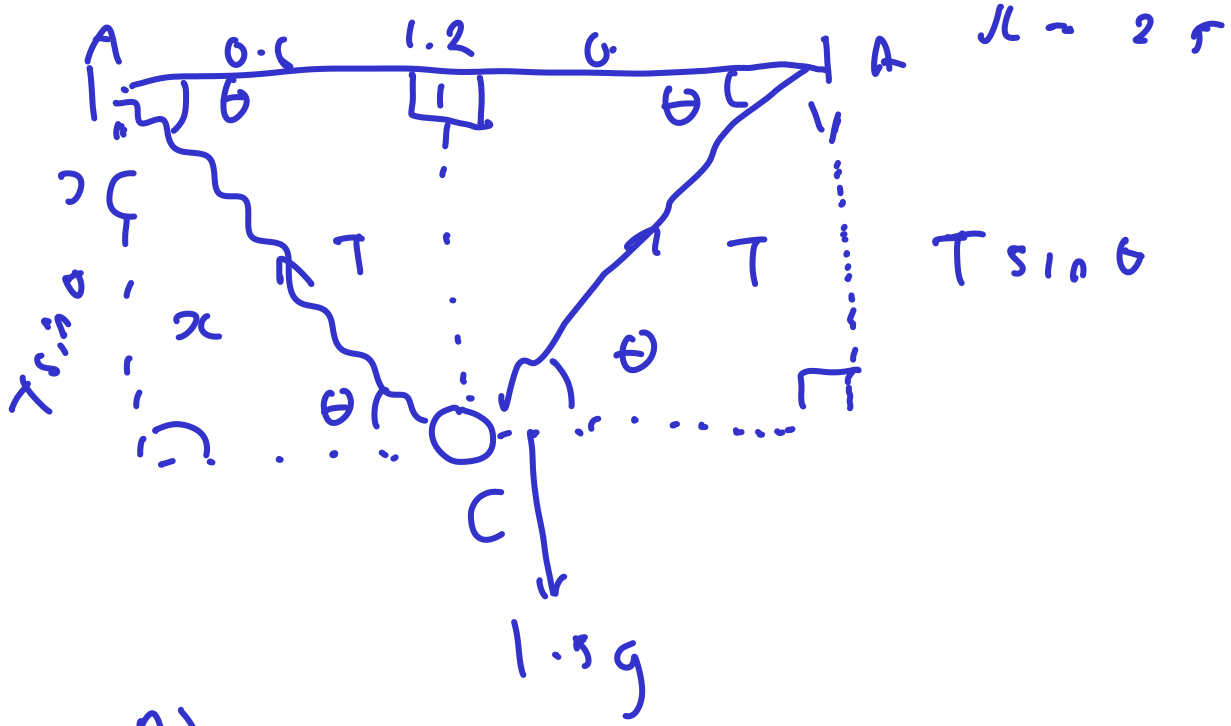
$\theta = 49.6°$ 3 sf

↳ the mark scheme is wrong

Two identical elastic springs, each of natural length $0.8\ m$ and modulus of elasticity $25\ N$, have one end attached to a particle of mass $1.5\ kg$. The other ends are attached to two fixed points X and Y on a horizontal line, where XY is $1.2\ m$. Find the total stretched length of one of the springs in equilibrium.

$l = 0.8$

$\lambda = 25$



$R(\uparrow)$

$2T\sin\theta = 1.5g$

Original length of spring = 0.8

length length of $xc = \frac{1}{\cos\theta}$

Extension $= \frac{1}{\cos\theta} - 0.8$

$$\therefore T = \frac{2.5\left(\frac{1}{\cos\theta} - 0.8\right)}{0.8}$$

$$T = 3\,1.25\left(\frac{1}{\cos\theta} - 0.8\right)$$

sub

$$2\left(31.25\left(\frac{1}{\cos\theta} - 0.8\right)\right)\sin\theta = 1.5(5.2)$$

$$62.5\sin\theta\left(\frac{1}{\cos\theta} - 0.8\right) = 1.5(5.2)$$

$$62.5\tan\theta - 50\sin\theta = 1$$

$$\theta = 34.53429^\circ$$

$\therefore$ elevation

$$= \frac{1}{\cos(AS)} = 1$$

$$= 1 + 0.8$$

$$= 1.8 //$$